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(57) Abstract:

Healthcare systems require continuous patient monitoring, rapid emergency response, and efficient management inside Intensive Care Units (ICU). Traditional monitoring methods often depend on manual observation, which can delay detection of critical health conditions. To overcome these limitations, this project proposes a Smart ICU Monitoring and Autonomous Emergency Response System using IoT. The system is designed to monitor multiple ICU beds in real time using advanced biomedical and environmental sensors. Each patient unit is equipped with sensors such as heart rate, ECG, SpO2, blood pressure, respiration, temperature, glucose, body movement, and bed occupancy sensors. These sensors continuously collect patient health data and send it to an ESP32-based controller. The data is transmitted wirelessly through NRF24L01, LORA, WiFi, or 5G communication modules to a central ICU gateway and cloud server. An Artificial Intelligence engine analyzes the incoming data to detect abnormalities, predict emergencies, and generate smart alerts before critical conditions occur. In case of danger, alerts are instantly sent to nurses, doctors, and family members through mobile applications, SMS, calls, and dashboard notifications. The system also supports voice nurse calls, robot assistant integration, automatic oxygen control, smart bed adjustment, and digital twin ICU monitoring for future hospital automation. The proposed project improves patient safety, reduces nurse workload, enables faster medical response, and enhances healthcare efficiency. It provides secure cloud data storage, historical health reports, and remote monitoring features. This smart ICU solution represents an advanced combination of IoT, AI, Cloud Computing, Robotics, and Healthcare Automation, making it highly useful for modern hospitals and future smart healthcare systems.

FORM 2

THE PATENTS ACT, 1970

(39 OF 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

TITLE OF THE INVENTION

AI POWERED SMART ICU MONITORING AND AUTONOMOUS EMERGENCY RESPONSE SYSTEM USING IOT

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PREAMBLE TO THE DESCRIPTION

The following specification describes the invention and the manner in which it is to be performed.

FIELD OF THE INVENTION

[0001] This invention relates to the field of healthcare monitoring systems, Internet of Things (IoT) technology, and Artificial Intelligence (AI). More specifically, it concerns an integrated, AI-powered system for real-time monitoring of multiple Intensive Care Unit (ICU) beds, autonomous emergency response, and remote healthcare management using advanced sensors and wireless communication. The invention further addresses the growing need for predictive healthcare analytics, automated clinical intervention, and seamless integration of robotics and digital twin technologies within critical care environments.

BACKGROUND OF THE INVENTION

[0002] Healthcare systems worldwide are under constant pressure to provide continuous, reliable, and rapid patient monitoring, particularly within Intensive Care Units (ICU) where patients are in critical condition. Timely detection of changes in vital signs and rapid emergency response are crucial for patient survival and recovery.

[0003] Key health parameters that directly affect patient outcomes in an ICU include heart rate, electrocardiogram (ECG) signals, blood oxygen saturation (SpO2), blood pressure, respiration rate, body temperature, glucose levels, body movement, and bed occupancy status. Among these, sudden changes in heart rate, ECG abnormalities, and drops in SpO2 are the most critical and require immediate medical attention.

[0004] Traditional monitoring methods rely on manual observation and conventional bedside monitors, which are labor-intensive and prone to error. Existing automated systems suffer from short communication range, lack of AI-driven prediction, inability to autonomously trigger emergency protocols (oxygen, bed adjustment), and no integration with robotics or digital twins.

[0005] Furthermore, most existing systems do not integrate multiple communication modalities (NRF24L01, LORA, Wi-Fi, 5G) for redundancy and reliability. They also lack comprehensive features like voice nurse calls, family notifications, and secure cloud-based historical health record keeping. There is no provision for an AI engine that can predict emergencies before they occur and initiate autonomous corrective actions.

[0006] Therefore, a comprehensive, intelligent, and autonomous system is required that can continuously monitor multiple health and environmental parameters across several ICU beds, transmit data reliably using diverse communication networks, predict emergencies using AI, and automatically initiate life-saving responses without continuous human intervention.

OBJECTS OF THE INVENTION

[0007] The primary object of this invention is to provide a smart ICU monitoring system that continuously monitors heart rate, ECG, SpO2, blood pressure, respiration, temperature, glucose, body movement, and bed occupancy in real time.

[0008] Another object is to implement an Artificial Intelligence engine that analyzes incoming patient data to detect abnormalities, predict potential medical emergencies, and generate smart alerts before critical conditions occur.

[0009] A further object is to provide autonomous emergency response actions, including automatic oxygen control and smart bed adjustment, based on AI-driven decisions.

[0010] An additional object is to integrate multiple wireless communication technologies (NRF24L01, LORA, Wi-Fi, 5G) to ensure reliable, redundant, and long-range data transmission from patient units to the central ICU gateway and cloud server.

[0011] Another object is to incorporate a voice nurse call system and robot assistant integration, allowing automated notifications and robotic assistance during emergencies.

[0012] A further object is to create a digital twin of the ICU for advanced monitoring, simulation, and future hospital automation.

[0013] Yet another object is to provide secure cloud data storage, historical health reports, and remote monitoring features accessible to doctors and family members via mobile applications and dashboards.

[0014] Another object is to send instantaneous emergency alerts through multiple channels, including mobile app notifications, SMS, and voice calls, to nurses, doctors, and family members.

[0015] A further object is to reduce nurse workload, enable faster medical response, and enhance overall healthcare efficiency in ICU settings.

SUMMARY OF THE INVENTION

[0016] This invention provides an integrated, AI-powered system and method for smart ICU monitoring and autonomous emergency response. The system enables continuous real-time monitoring of multiple critical health and environmental parameters across several ICU beds, automatic AI-driven emergency responses, and remote data access via cloud and mobile applications.

[0017] The system comprises multiple patient sensor nodes, each with heart rate, ECG, SpO2, blood pressure, respiration, temperature, glucose, body movement, and bed occupancy sensors interfaced to an ESP32 microcontroller. Power comes from a hospital supply with battery backup. Communication uses NRF24L01, LORA, Wi-Fi, or 5G. An AI engine, an emergency response unit (oxygen valve and smart bed), voice nurse call, and robot interface are also included.

[0018] Sensor data will be continuously collected by the ESP32 controller and transmitted via the most suitable available communication channel (NRF24L01, LORA, Wi-Fi, or 5G) to a central ICU gateway. The gateway forwards the data to a cloud server. The AI engine on the cloud or at the edge analyzes the data in real time to detect patterns, predict emergencies (e.g., cardiac arrest, respiratory failure), and generate smart alerts.

[0019] When a critical condition predicted or detected, the system automatically initiates a series of emergency responses. This may include adjusting the smart bed to a safe position, activating automated oxygen supply, triggering a voice nurse call, and sending alerts to medical staff. The system can also integrate with a robot assistant to deliver supplies or assist medical staff.

[0020] All patient data -- including vital signs, movement, occupancy, AI-generated alerts, and emergency response actions -- are stored securely in the cloud. Authorized users (doctors, nurses, family members) can access real-time data, receive alerts, and view historical reports through a mobile application or web dashboard. The system also maintains a digital twin of the ICU environment for advanced monitoring and simulation.

BRIEF DESCRIPTION OF DRAWINGS

[0021] The invention is further described with reference to the accompanying drawings:

[0022] FIG. 1 shows a block diagram of the overall system architecture for the AI-powered Smart ICU monitoring system, including multiple patient beds, communication gateways, cloud AI engine, and user interfaces.

[0023] FIG. 2 shows a detailed block diagram of a patient sensor node, including all biomedical and environmental sensors (heart rate, ECG, SpO2, blood pressure, respiration, temperature, glucose, body movement, bed occupancy), ESP32 controller, multiple communication modules (NRF24L01, LORA, Wi-Fi, 5G), and emergency response actuators.

[0024] FIG. 3 and FIG. 4 shows block diagrams of the central ICU gateway with communication interfaces with receiver unit.

[0025] FIG. 5 shows a flowchart of the operational logic of the system, including continuous sensor reading, AI-based abnormality detection, emergency prediction, autonomous response activation, alert generation, and data logging.

DETAILED DESCRIPTION OF THE INVENTION

[0026] The following detailed description explains the invention with reference to figures 1 through 5. It will be understood that while specific embodiments are described, the underlying principles might be implemented in various forms.

[0027] FIG. 1 provides an overview of the complete system architecture. Multiple ICU beds are monitored using individual patient sensor nodes. As shown in FIG. 5 N representative nodes: NodeA at Bed 1, NodeB at Bed 2, NodeC at Bed 3, and NodeN at Bed N. Each node contains a full suite of biomedical and environmental sensors as well as emergency response actuators.

[0028] Each node communicates wirelessly using one or more of the available technologies: NRF24L01 for short-range low-power communication, LORA for long-range low-power transmission, Wi-Fi for high-bandwidth local networking, and 5G for ultra-reliable, low-latency cellular communication. The system intelligently selects the best available channel.

[0029] A Central ICU Gateway is placed within the ICU. It contains complementary receivers for NRF24L01, LORA, Wi-Fi, and 5G, along with a powerful processor. The gateway collects data packets from all patient nodes and forwards them to the cloud server.

[0030] The Cloud Server hosts the AI Engine, which processes incoming data, stores it in a secure database, runs predictive algorithms, and makes data available through two interfaces: a Mobile Application for doctors, nurses, and family members, and a Web Dashboard for detailed analysis, digital twin visualization, and historical trends. The system also includes an Emergency Alert System to notify staff via SMS, voice calls, and app notifications.

[0031] FIG. 2 details the internal components of a patient sensor node (200). The node is powered by a hospital-grade power supply with an uninterruptible battery backup (203).

[0032] The core of the node is an ESP32 microcontroller (201). This powerful, low-energy microcontroller with integrated Wi-Fi and Bluetooth manages all operations: reading sensors, executing control logic, managing communication, activating emergency responses, and handling voice calls.

[0033] The following sensors are connected to the microcontroller:

- Heart rate sensor -- measures patient pulse.
- ECG sensor -- records electrical activity of the heart.
- SpO2 sensor -- measures blood oxygen saturation.
- Blood pressure sensor -- measures systolic and diastolic pressure.
- Respiration sensor -- measures breathing rate.
- Temperature sensor -- measures patient body temperature.
- Glucose sensor -- measures blood glucose levels.
- Body movement sensor -- detects patient movement or seizures.
- Bed occupancy sensor -- detects if the bed is occupied.

[0034] A multi-communication module (206) integrates NRF24L01, LORA, Wi-Fi, and 5G interfaces, allowing the ESP32 to select the optimal communication method based on network conditions and data urgency.

[0035] For autonomous emergency response, the node includes:

- An automatic oxygen control valve that can increase or decrease oxygen flow based on AI commands.

- A smart bed adjustment mechanism that can change bed position (e.g., raising the head) during emergencies.
- An interface for robot assistant integration to summon or communicate with a hospital robot.

[0036] The node also includes a voice nurse call unit allowing the patient or the AI to directly call the nursing station.

[0037] FIG. 3 illustrates the central ICU gateway (300). It includes:

- Multiple receiver modules for NRF24L01, LORA, Wi-Fi, and 5G and a stable power supply
- A high-performance processor for data aggregation and local edge AI processing.
- An Ethernet or fibre optic connection to the internet for cloud communication.
- A local buzzer and LED for audio-visual alerts within the ICU.

[0038] FIG. 5 shows the operational flowchart of the system. The process begins with start/power-on followed by initialization (402) of all sensors, communication modules, and AI engine. The system then enters continuous monitoring mode (403) collecting data from all sensors at regular intervals.

[0039] The collected data is fed into the AI Engine which analyzes the data stream for abnormalities, trends, and early signs of emergencies. The AI uses trained models to predict conditions such as cardiac arrest, respiratory distress, or sepsis.

[0040] At decision point the AI assesses the patient's condition:

- If all parameters are normal, the system simply logs the data and continues monitoring.

- If an abnormality is detected but no immediate emergency the system generates a warning alert to the nursing dashboard.
- If a critical emergency is predicted or detected the system initiates autonomous emergency response This includes activating oxygen control, adjusting the smart bed, triggering a voice nurse call, and sending instant alerts to doctors, nurses, and family members via SMS, app, and voice call.

[0041] Simultaneously, all data -- including raw sensor readings, AI analysis, and emergency actions -- is transmitted via the gateway to the cloud server for storage and digital twin updating. The system continuously cycles ensuring no delay in emergency detection and response.

[0042] The cloud server stores all data securely, generates historical health reports, and updates the digital twin of the ICU. The mobile application allows remote monitoring, alert management, and system configuration.

CLAIMS

1. A smart ICU monitoring and autonomous emergency response system using IoT, comprising:

a. A plurality of patient sensor nodes configured for deployment in different ICU beds, each patient sensor node comprising:

  i. A plurality of biomedical and environmental sensors including a heart rate sensor, an ECG sensor, an SpO2 sensor, a blood pressure sensor, a respiration sensor, a temperature sensor, a glucose sensor, body movement sensor, and a bed occupancy sensor;

  ii. An ESP32-based microcontroller operatively interfaced with said sensors;

  iii. A multi-communication module integrating NRF24L01, LORA, Wi-Fi, and 5G for redundant and reliable wireless data transmission;

  iv. An autonomous emergency response unit comprising an automatic oxygen control valve and a smart bed adjustment mechanism;

  v. A voice nurse call unit and a central ICU gateway configured to receive data from said plurality of patient sensor nodes;

b. A cloud server comprising an Artificial Intelligence engine for data analysis, abnormality detection, and emergency prediction.

c. A mobile application and web dashboard configured to display real-time patient data, AI-generated alerts, and historical reports.

2. The system as claimed in claim 1, wherein the Artificial Intelligence engine is configured to analyse incoming patient data to predict medical emergencies including cardiac arrest and respiratory failure before critical conditions occur.

3. The system as claimed in claim 1, wherein the ESP32-based microcontroller is configured to automatically activate the autonomous emergency response unit, including increasing oxygen flow and adjusting bed position, based on commands received from the AI engine.

4. The system as claimed in claim 1, wherein the system transmits data via the most suitable available communication channel selected from NRF24L01, LORA, Wi-Fi, and 5G, ensuring communication redundancy.

5. The system as claimed in claim 1, wherein the voice nurse call unit is configured for automated calling to a nursing station upon detection of a critical event by the AI engine.

6. The system as claimed in claim 1, wherein the cloud server maintains a digital twin of the ICU environment for advanced monitoring, simulation, and hospital automation.

7. The system as claimed in claim 1, wherein the mobile application is configured to send instantaneous emergency alerts to nurses, doctors, and family members through push notifications, SMS, and voice calls.

8. A method for smart ICU monitoring and autonomous emergency response using the system of any preceding claim, comprising the steps of:

a. Deploying a plurality of patient sensor nodes in different ICU beds;

b. Continuously monitoring, at each node, heart rate, ECG, SpO2, blood pressure, respiration, temperature, glucose, body movement, and bed occupancy;

c. Transmitting all sensor data wirelessly using NRF24L01, LORA, Wi-Fi, or 5G to a central ICU gateway and then to a cloud server;

d. Analysing, by an AI engine on the cloud server or edge gateway, the incoming data to detect abnormalities and predict emergencies;

e. Upon prediction or detection of a critical condition, automatically activating the autonomous emergency response unit to control oxygen supply and adjust the smart bed;

f. Triggering a voice nurse call and sending instant alerts to medical staff and family members via mobile application, SMS, and voice call;

g. Securely storing all patient data, AI analysis results, and emergency response logs in the cloud; and
h. Updating a digital twin of the ICU for future automation and simulation.

9. The method as claimed in claim 8, wherein the step of analyzing includes using trained AI models to predict conditions such as cardiac arrest, respiratory distress, or sepsis before clinical symptoms become apparent.

10. The method as claimed in claim 8, further including the step of integrating a robot assistant to deliver supplies or assist medical staff during an emergency.

ABSTRACT

AI POWERED SMART ICU MONITORING AND AUTONOMOUS EMERGENCY RESPONSE SYSTEM USING IOT

Healthcare systems require continuous patient monitoring, rapid emergency response, and efficient management inside Intensive Care Units (ICU). Traditional monitoring methods often depend on manual observation, which can delay detection of critical health conditions. To overcome these limitations, this project proposes a Smart ICU Monitoring and Autonomous Emergency Response System using IoT.

The system is designed to monitor multiple ICU beds in real time using advanced biomedical and environmental sensors. Each patient unit is equipped with sensors such as heart rate, ECG, SpO2, blood pressure, respiration, temperature, glucose, body movement, and bed occupancy sensors. These sensors continuously collect patient health data and send it to an ESP32-based controller. The data is transmitted wirelessly through NRF24L01, LORA, WiFi, or 5G communication modules to a central ICU gateway and cloud server.

An Artificial Intelligence engine analyzes the incoming data to detect abnormalities, predict emergencies, and generate smart alerts before critical conditions occur. In case of danger, alerts are instantly sent to nurses, doctors, and family members through mobile applications, SMS, calls, and dashboard notifications. The system also supports voice nurse calls, robot assistant integration, automatic oxygen control, smart bed adjustment, and digital twin ICU monitoring for future hospital automation.

The proposed project improves patient safety, reduces nurse workload, enables faster medical response, and enhances healthcare efficiency. It provides secure cloud data storage, historical health reports, and remote monitoring features. This smart ICU solution represents an

advanced combination of IoT, AI, Cloud Computing, Robotics, and Healthcare Automation, making it highly useful for modern hospitals and future smart healthcare systems.